
a Input (one gene at a time)
s_2
s_1
Type 1 Type 2 Type 3 Type 4
y Expression n × 1
X Covariates n × p
S Spatial coordinates n × 2
Π Cell-type composition n × K
b Construct kernels
K_s
Spatial kernel
from coordinates S
K_c
Cell-type kernel
from composition Π
$K_{ct} = K_s \odot K_c$
Interaction kernel
Hadamard product
c SIGNAL linear mixed model
$y = X\beta + b_s + b_c + b_{ct} + \varepsilon$
fixed effects
spatial domain
cell-type composition
cell-type-specific spatial
noise
$b_s \sim MVN(0, \sigma_s^2 K_s)$
$b_c \sim MVN(0, \sigma_c^2 K_c)$
$b_{ct} \sim MVN(0, \sigma_{ct}^2 K_{ct})$
$var(y) = \sigma_s^2 K_s + \sigma_c^2 K_c + \sigma_{ct}^2 K_{ct} + \sigma_0^2 I_n$
Each variance component σ² measures one source of spatial variation.
d Stage 1 · overall test
$H_0: \sigma_s^2 = \sigma_c^2 = \sigma_{ct}^2 = 0$
$H_1:$ at least one component > 0
p-values from 11 candidate spatial kernels
1 2 3 4 5 6 7 8 9 10 11
Cauchy combination → one p-value
FDR control across genes (Benjamini–Yekutieli)
utSVGs
e Stage 2 · variance decomposition
For each utSVG, test each component separately
$H_0: \sigma_s^2 = 0$ $H_1: \sigma_s^2 > 0$
$H_0: \sigma_c^2 = 0$ $H_1: \sigma_c^2 > 0$
$H_0: \sigma_{ct}^2 = 0$ $H_1: \sigma_{ct}^2 > 0$
sSVG
spatial-domain driven
cSVG
cell-type-composition driven
ctSVG
cell-type-specific
f Downstream analyses
Cell-type-k-specific SVGs
Cell-type-specific gene-set enrichment
$-\log_{10} P$
Spatial domain detection